

Lecture Notes in Civil Engineering

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Real-Time Application of Computer Vision for Displacement Monitoring of Industrial Steel Frame Structure

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Abstract. Displacement is a critical factor in evaluating the structural integrity of various structures, especially steel structures. However, traditional displacement measurement methods are costly and often impractical for real-time applications. In this study, we proposed a computer vision system using low-cost surveillance cameras designed specifically for real-time displacement measurement of industrial steel frame structures subjected to wind loads. The system was deployed to monitor displacement due to wind load at the facility site over a period of 24 days. The results demonstrate that the system can perform non-contact, real-time displacement monitoring while also being low-cost and easy to install.

Keywords: Computer vision · Displacement · Steel structure · Wind velocity · Structural health monitoring · Surveillance cameras

1 Introduction

Structural health monitoring (SHM) is crucial for ensuring the safety and efficiency of structures. This system detects damage and deterioration early, allowing for timely maintenance and repair. SHM involves collecting data from sensors that measure stress, displacement, temperature, humidity, and other environmental factors. This data helps assess the current condition of the structure and predict its load-bearing capacity and service life. Consequently, managers can optimize maintenance costs and enhance user safety.

Recognizing the importance of SHM, several decrees and circulars have been issued [1–3], mandating monitoring throughout the lifespan of construction works. This trend highlights the need for integrating SHM systems in future construction projects, especially large-scale structures such as bridges, high-rise buildings, and industrial steel structures. However, integration of SHM faces significant challenges, primarily due to high costs and the complexity of installation processes. Traditional methods, such as accelerometers [4] and laser sensors [5], are costly in terms of initial investment, operation, maintenance, data collection, storage, and analysis.

Over the last few decades, significant progress has been made in using computer vision for monitoring the displacement of structures. Lee developed a remote system to measure the dynamic displacement of a bridge [6]. Feng introduced a real-time, non-contact structural displacement measurement system utilizing an upsampling inter-correlation algorithm [7]. Dong applied a scale-invariant feature transform (SIFT) for feature matching to monitor the displacement of a two-span bridge structure, and railroad bridges [8]. Huang developed a computer vision system for deflection monitoring steel structures subjected to prolonged loading [9].

In this study, we proposed a computer vision system using low-cost surveillance cameras designed specifically for real-time displacement measurement of industrial steel frame structures subjected to wind loads. The system was deployed to monitor displacement due to wind load at the facility site over a period of 24 days. The results demonstrate that the system can perform non-contact, real-time displacement monitoring while also being low-cost and easy to install.

2 System Configuration

2.1 Hardware Configuration

Figure 1 illustrates the hardware configuration of a real-time, computer vision-based structural deformation monitoring system. The setup includes a camera and lens for image acquisition. In this study, a Dahua DAS-BN5461T-ZS surveillance camera with a 1/2.9" CMOS sensor and a resolution of 2688×1520 at one frame per second (fps) was utilized. The camera features an integrated lens with a variable focal length ranging from 2.7 mm to 13.5 mm. Video data is transmitted via the internet to a Network Video Recorder (NVR) using a Ruijie RG-EG105G-V2 router, which has a bandwidth of 600 Mbps. The NVR employed in this system is a Dahua NVR4116HS-4KS2/L, equipped with 2TB of storage. Finally, the video is transferred to a workstation computer (Dell Precision Tower 5820) for image processing.

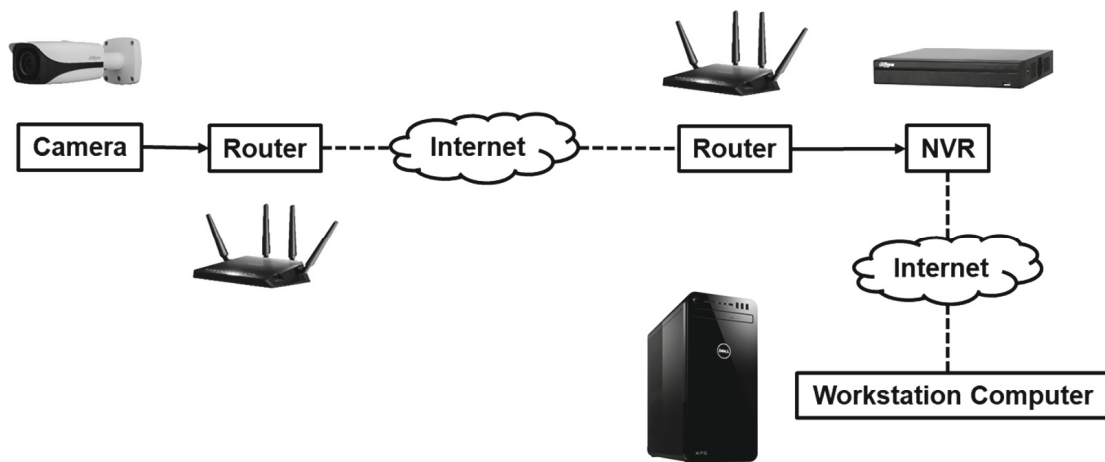


Fig. 1. Hardware configuration of a real-time computer vision-based structural deformation monitoring system.

2.2 Software Configuration

Figure 2 depicts the software configuration of the system. Initially, the first frame of the video is extracted, followed by the selection of a region of interest (ROI) that encompasses the area required for displacement monitoring. Within this ROI, feature points are identified using the Speeded Up Robust Features (SURF) algorithm [10]. In subsequent frames, the same ROI is used to detect feature points. The differences in pixel coordinates in the x and y directions are then calculated. Finally, the real-world displacement is determined by multiplying the differences in pixel coordinates with a scale factor, which is the ratio between the actual distance (D) and the distance in pixel coordinates (p).

Since multiple feature points are identified, the actual displacement is calculated as the average displacement of all feature points. This is because, within a small ROI area, the displacement is assumed to be the same for each feature point.

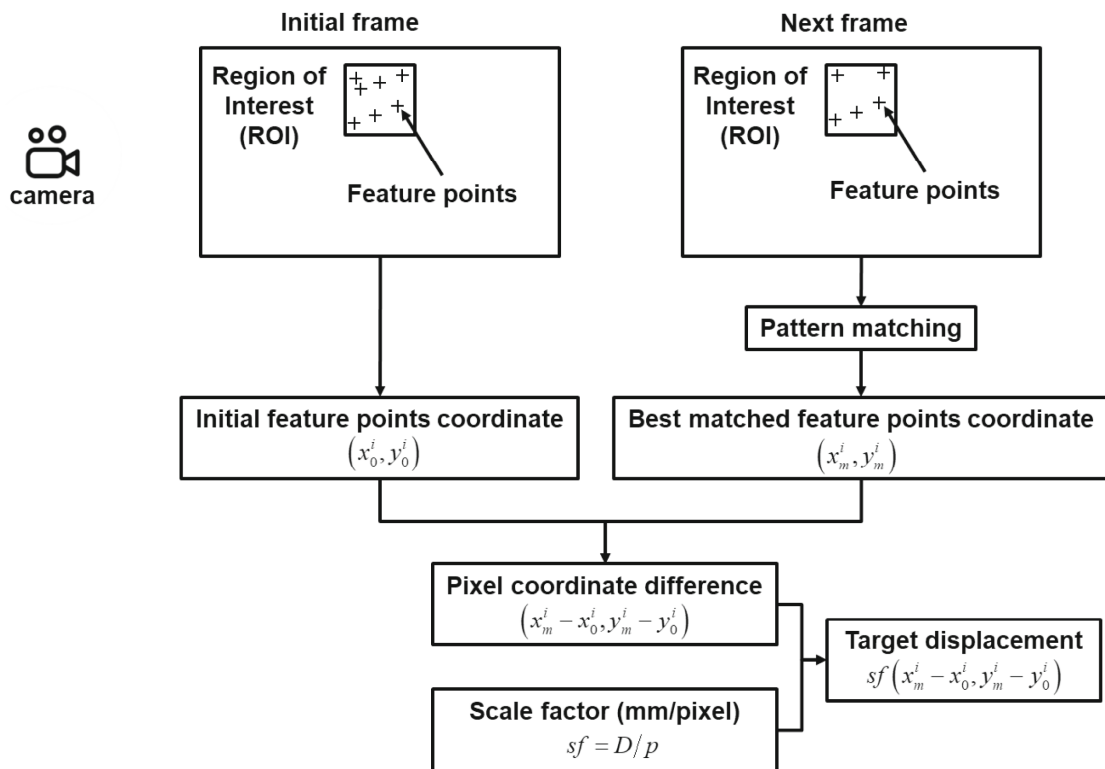


Fig. 2. Software configuration of the real-time computer vision-based structural deformation monitoring system.

3 Real-World Application of the Real-Time Computer Vision-Based Structural Deformation Monitoring System.

3.1 System Installation

Figure 3 demonstrates the real-world application of the real-time computer vision-based structural deformation monitoring system on an industrial steel frame structure. The factory is located in Suoi Dau Industrial Park, Cam Lam Commune, Nha Trang City.

It consists of two workshops: Workshop 1, where only a portion of the roof is covered with solar panels, and Workshop 2, where the roof is entirely covered with solar panels. The factory owner has requested an assessment of the impact of the newly installed solar panels on the roof, specifically to investigate the displacement caused by wind and determine if it falls within the allowable range specified by Vietnam Standard 5575:2012 for steel structures.

Figure 4 shows the installation of the camera system. The camera was installed on the gable wall of the workshops. Since the gable wall is made of brick, it is reasonable to assume that its vertical and horizontal displacements are minimal compared to the displacement at the middle of the investigated frame. The distance from the camera to the frame is 12 m. The process of installing the whole system, including the camera and video data transfer system, was completed only one working day.

A wind velocity sensor (RS485 Modbus RTU ES-WS-02) was installed on top of the office building near the factory (Fig. 5). The data from the wind velocity sensor was captured using a PLC system and uploaded to the internet in real-time via a web app.



Fig. 3. Image of the factory with installed real-time computer vision-based structural deformation monitoring system.



Fig. 4. Camera Installation and Monitoring Frame: The distance from the camera to the frame is 12 m. No artificial target was installed on the frame.

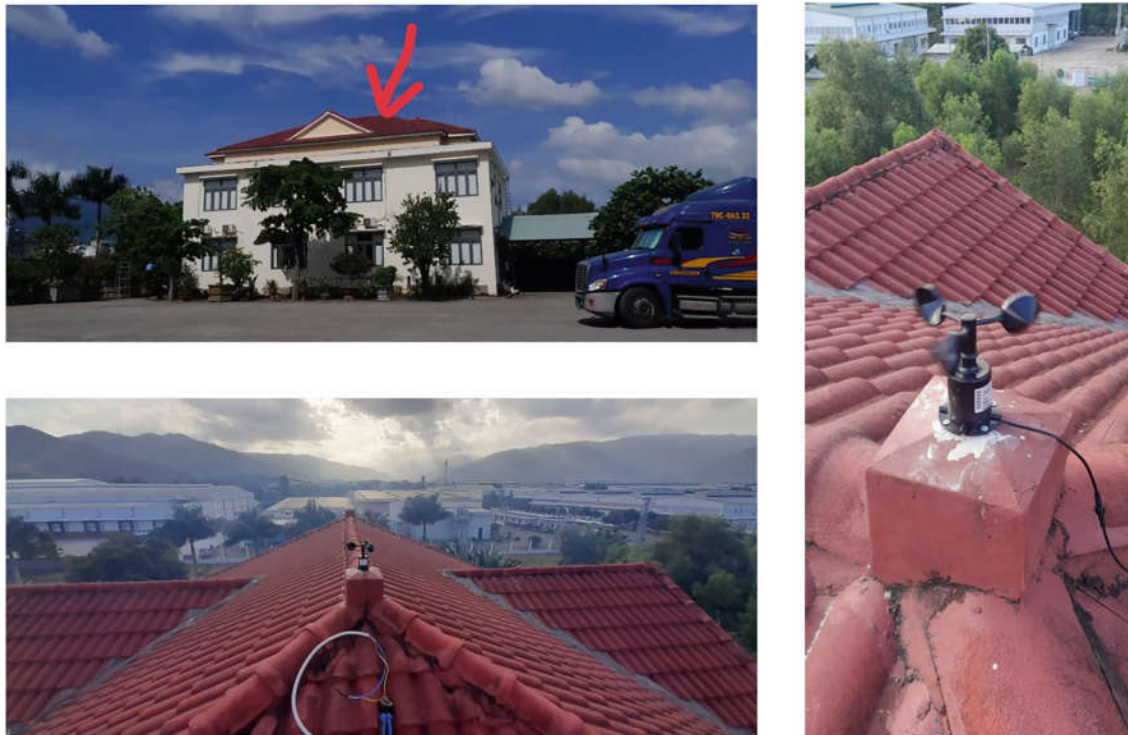


Fig. 5. Location of installed wind velocity sensor and image of the wind velocity sensor at the top of the office building.

3.2 Displacement Monitoring Results

Figure 6 shows the MATLAB application used for displaying displacement monitoring results of steel frame structures. The user begins by selecting the camera channel, either CH1 or CH2, corresponding to two workshops and the date of the video data. The image displayed below the date shows the video frame with the selected ROI. Yellow crosses within the ROI indicate feature points detected using the SURF algorithm. The displacement data is presented on the right side of the app, with the upper graph showing vertical displacement and the lower graph showing horizontal displacement.

During the monitoring period, it was found that the maximum vertical displacement of the steel frame structures in Workshop 1 and Workshop 2 was 18.1 mm and 19.3 mm, respectively. The maximum horizontal displacement of the steel frame structures in Workshop 1 and Workshop 2 was 11.2 mm and 15.5 mm, respectively. Therefore, it is determined that displacement falls within the allowable range specified by Vietnam Standard 5575:2012 for steel structures.

Figure 7 displays the wind velocity graph from January 26 to February 16. It is noted that data is unavailable on certain days due to sensor malfunction. A well-correlated relationship between wind speed and the displacement of the frame is observed, as shown in Fig. 8.

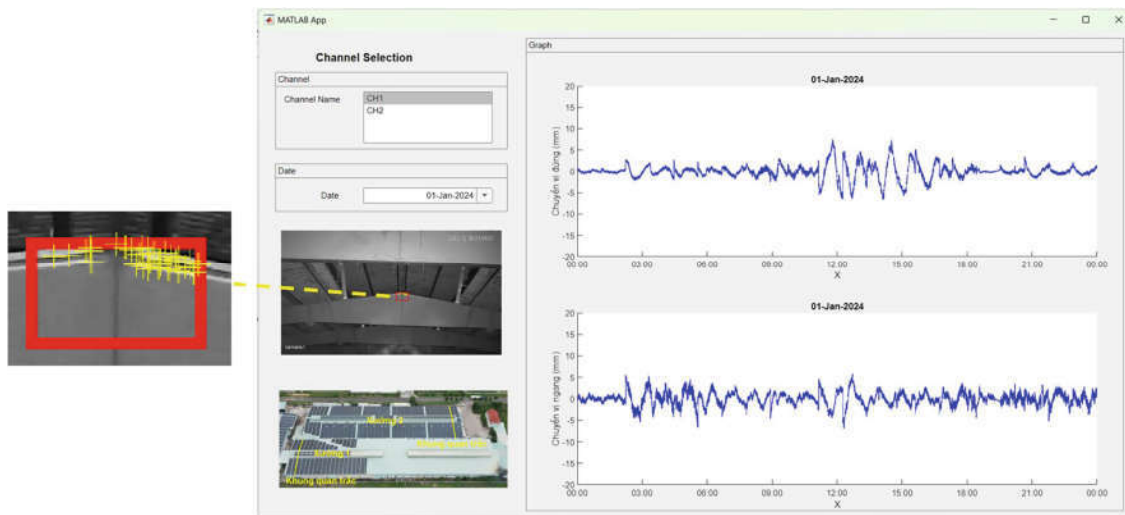


Fig. 6. MATLAB app displaying displacement monitoring results of steel frame structures. The upper graph shows vertical displacement, and the lower graph shows horizontal displacement of the frame on 01 January 2024.

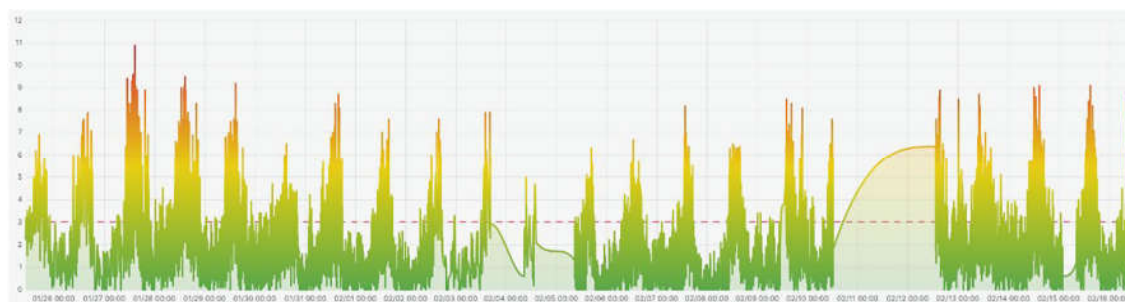


Fig. 7. Wind velocity graph from January 26 to February 16. *Note* Data is unavailable on certain days due to power shortages.

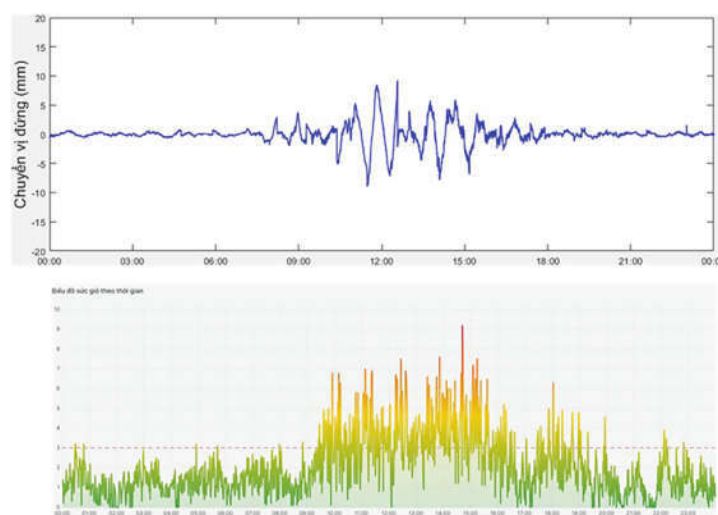


Fig. 8. Comparison of vertical displacement of the frame structure and wind velocity on a typical day.

4 Conclusions

This study has successfully demonstrated the application of a real-time computer vision-based system for displacement monitoring of industrial steel frame structures. Utilizing low-cost surveillance cameras, the system effectively monitored the displacement caused by wind loads over a period of 24 days. The results confirmed that the displacements of the workshop remained within the allowable range specified by Vietnam Standard 5575:2012, thereby ensuring structural safety.

By applying advanced computer vision technology, this approach offers a comprehensive solution for continuous structural integrity assessments. The system's cost-effectiveness, ease of installation, and real-time data acquisition capabilities highlight its potential for widespread adoption in various structural monitoring applications.

The insights gained from this study emphasize the importance of innovative monitoring techniques in maintaining the safety and longevity of critical infrastructure. Continued research and development in this field are essential to enhance further the reliability and efficiency of structural health monitoring systems.

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